Sine and cosine functions

1

a $\frac{dy}{dx} = -3\sin 3x$

b $\frac{dy}{dx} = 7\cos 7x$

c $\frac{dy}{dx} = 3\sin x$

d $\frac{dy}{dx} = \cos x - 3\sin x$

e $\frac{dy}{dx} = -3\cos x$

f $\frac{dy}{dx} = 6x^5\cos\left(x^6 + 5\right)$

g $\frac{dy}{dx} = 20x^4\cos\left(4x^5 + 3\right)$

h $\frac{dy}{dx} = 3 + 5\cos 5x$

i $\frac{dy}{dx} = 4x^3\cos\left(x^4 + 3\right)$

j $\frac{dy}{dx} = -6x\sin\left(3x^2\right)$

k $\frac{dy}{dx} = \frac{1}{3}\cos\left(\frac{x}{3}\right) + 2\sin x$

l $\frac{dy}{dx} = 5x^4\cos\left(x^5\right)$

m $\frac{dy}{dx} = \cos\left(\frac{t}{4}\right) - 12\sin 4t + 4t^3$

2

a $\frac{dy}{dx} = 4\left(\cos x - x\sin x\right)$

b $\frac{dy}{dx} = \cos 2x$

c $\frac{dy}{dx} = 18\sin^5\left(3x\right)\cos 3x$

d $\frac{dy}{dx} = -\sin 4x\sin x + 8\cos 4x + 4\cos x\cos 4x$

e $\frac{dy}{dx} = -\frac{2\sin 4x}{\sqrt{\cos 4x}}$

f $\frac{dy}{dx} = \sin 2x$

g $\frac{dy}{dx} = 5\sin 10x$

h $\frac{dy}{dx} = \frac{(x-8)\cos x - \sin x}{(x-8)^2}$

i $\frac{dy}{dx} = -5x^4\sin 5x - 15\sin 5x + 4x^3\cos 5x$

j $\frac{dy}{dx} = 6\cos 4x\cos 6x - 4\sin 6x\sin 4x$

k $\frac{dy}{dx} = -2x^{\frac{7}{4}}\sin 2x + \frac{7}{4}x^{\frac{3}{4}}\cos 2x$

l $\frac{dy}{dx} = -12\sin 4x + 30\cos 5x$

m $\frac{dy}{dx} = \frac{3\left(\sin x - x\cos x\right)}{\sin^2\left(x\right)}$

n $\frac{dy}{dx} = 4x^3\sin\left(\frac{x}{2}\right) + \frac{x^4\cos\left(\frac{x}{2}\right)}{2}$

o $\frac{dy}{dx} = x^2\left(3\cos x - x\sin x\right)$

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p $\dfrac{dy}{dx} = 20\cos 10x$

Applications

3 $y' = \dfrac{5x\cos 5x - \sin 5x}{x^2}$

4 $\dfrac{dy}{dx} = \dfrac{1}{\cos^2(x)}$

5 $\sin^2 x + \cos^2 x = 1$ for all x. Therefore function y is a constant function for all x and has a gradient of 0.

6

a D **b** B **c** A, C, E

7

a $\dfrac{d}{dx}(\sin x) = 1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} - \dfrac{x^6}{6!} + \dfrac{x^8}{8!} - \ldots$

b $\dfrac{d}{dx}(\cos x) = -x + \dfrac{x^3}{3!} - \dfrac{x^5}{5!} + \dfrac{x^7}{7!} - \dfrac{x^9}{9!} + \ldots$

c $\dfrac{d}{dx}(\sin x) = \cos x, \dfrac{d}{dx}(\cos x) = -\sin x$

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